

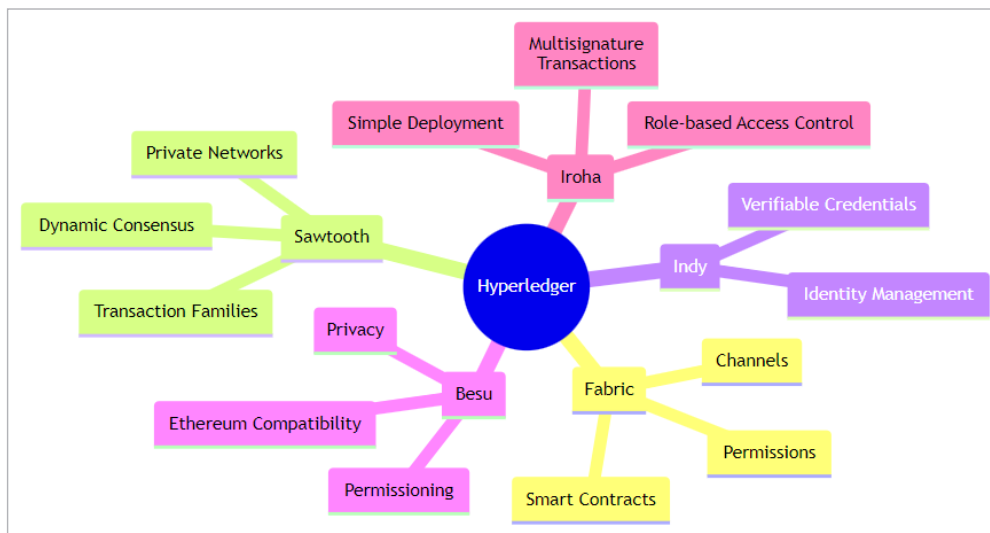


Figure 2: Hyperledger framework

- **Scalability and efficiency:** Hyperledger's architecture emphasises scalability, security, and efficiency for enterprise needs.

Hyperledger frameworks and tools

Hyperledger provides several frameworks and tools for building blockchain-based solutions. Here are some of the key frameworks and tools offered under the Hyperledger umbrella.

systems, Hyperledger enables seamless communication and data exchange across disparate platforms, enhancing collaboration and compatibility.

Industry-specific solutions:

Hyperledger provides frameworks and tools tailored to specific industries, such as finance, supply chain, healthcare, and more, addressing industry-specific challenges and requirements with customised blockchain solutions.

Innovation: It fosters innovation through open source collaboration, allowing developers and organisations to contribute to and leverage cutting-edge blockchain technology to drive digital transformation and create new business opportunities.

Compliance and regulation:

Hyperledger facilitates compliance with regulatory requirements by providing features such as permissioned networks, privacy controls, and audit trails, ensuring that businesses can meet legal and regulatory obligations while leveraging blockchain technology.

How does Hyperledger work?

Hyperledger works by providing a set of frameworks and tools that enable the creation, deployment, and management of enterprise-grade blockchain networks. Each

Hyperledger framework and tool addresses specific needs, allowing organisations to choose the components that best fit their use case.

- **Permissioned network:** Hyperledger operates on a permissioned network where participants are known and authenticated.
- **Consensus mechanism:** It uses modular consensus mechanisms, allowing organisations to choose the protocol that suits their needs.
- **Smart contracts:** Smart contracts, called chaincode, define transaction logic and rules.
- **Transaction processing:** Transactions are proposed by clients, endorsed by peers, ordered into blocks, and then committed to the ledger.
- **Ledger management:** Hyperledger maintains a world state representing the current blockchain state and a transaction log recording all transaction history.
- **Privacy and access control:** Features like channels and private data collections ensure privacy and selective sharing.
- **Identity management:** Solutions authenticate and authorise network participants for secure interactions and data access.

Hyperledger Fabric

- Modular blockchain framework designed for enterprise use.
- Supports permissioned networks with a focus on confidentiality, scalability, and flexibility.
- Features pluggable consensus protocols, smart contract support (chaincode), and a robust permission model.
- Suitable for a wide range of applications, including supply chain management, trade finance, and healthcare.

Hyperledger Sawtooth

- Modular platform for building, deploying, and running distributed ledgers.
- Utilises a unique consensus algorithm called Proof of Elapsed Time (PoET) and supports Ethereum-based smart contracts (Solidity).
- Offers scalability, security, and flexibility for various use cases such as supply chain tracking, digital identity, and financial services.

Hyperledger Besu

- Ethereum client built on Java, enabling private and public